

SOC IP Governance Automation System

Detailed Technical Report with Flow Diagrams and Visual Explanations

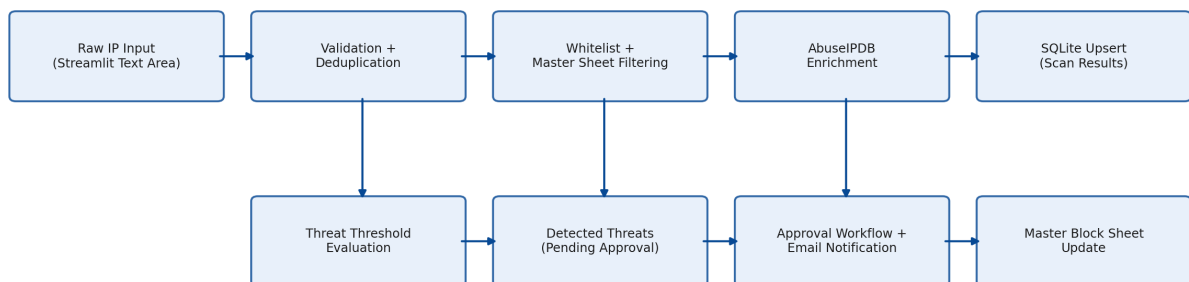
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Classification: Confidential

1. System Overview

This SOC IP Governance Automation platform standardizes passive SOC IP handling by combining validation, enrichment, approval, and documentation in one guided workflow. It reduces repetitive manual effort and improves consistency, traceability, and operational control.

2. End-to-End Flow Diagram

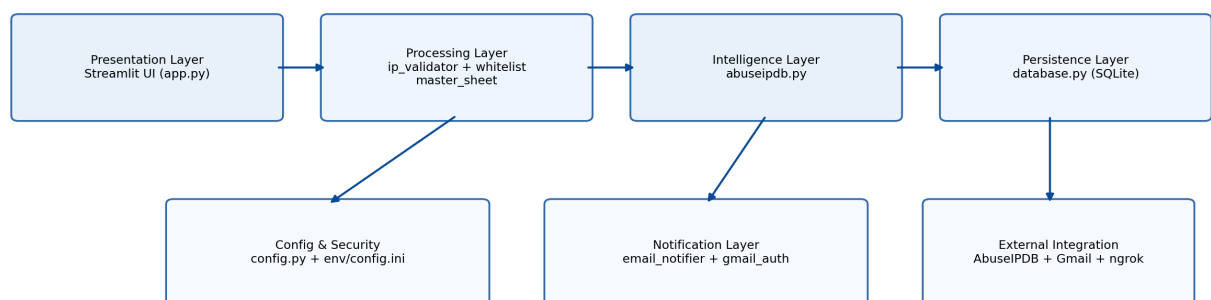
The following flow diagram shows how raw input moves through validation, threat intelligence enrichment, approval gates, and final master-sheet governance updates.



Passive SOC governance pipeline: validate → enrich → approve → document

3. Architecture Diagram

The architecture is modular to keep responsibilities isolated (UI, processing, integrations, persistence, notifications), making the system easier to maintain and extend.



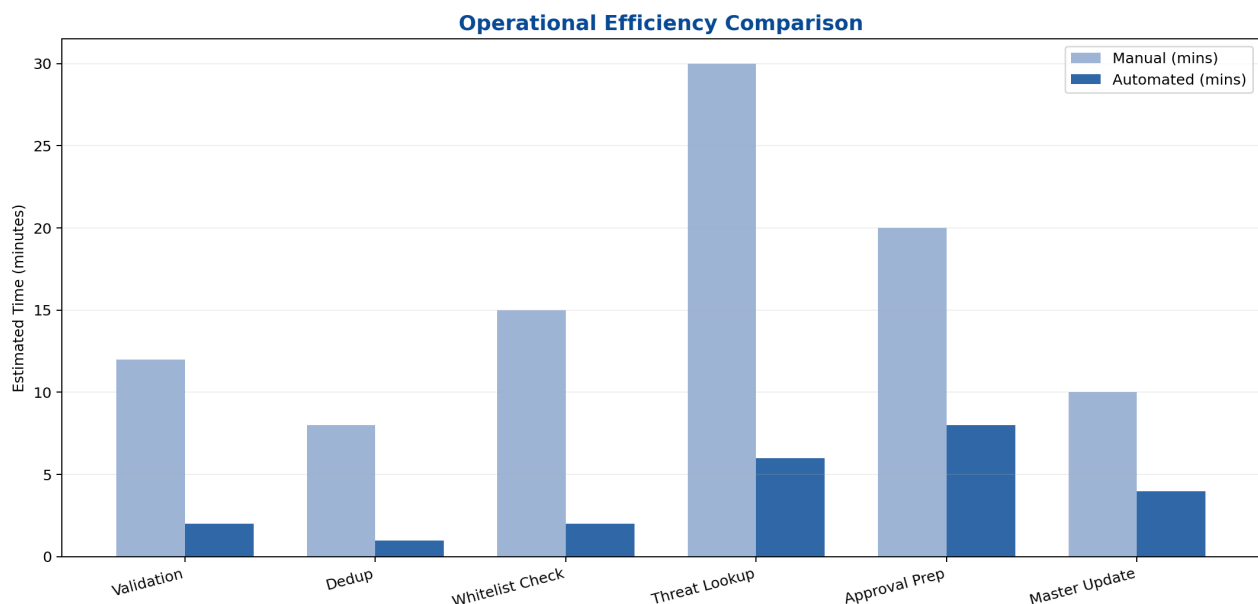
Modular architecture: UI orchestration with isolated integration and persistence responsibilities

4. Key Features

- Strict IPv4/IPv6 input validation and invalid entry isolation
- Duplicate removal while preserving original input order
- Whitelist filtering with individual IP and CIDR support
- Already-blocked filtering from authoritative master sheet
- AbuseIPDB enrichment with retry/backoff and 429 handling
- SQLite persistence for scan results and detected threats
- Approval workflow with status updates and email notifications
- Single-command launch and static ngrok public URL support

5. Efficiency Visualization

The chart below compares estimated analyst effort for manual handling versus current automated workflow. Actual timings vary with volume and API response time, but the reduction trend is significant.



6. Pros and Cons

Pros	Cons
High reduction in repetitive SOC effort	Depends on external services (AbuseIPDB/ngrok/Gmail)
Consistent governance and auditability via SQLite + structured CSV	Static CSV domain may conflict if another session is active
Improved decision context (score, country, ISP, path)	Passive mode: no direct firewall automation
Simple onboarding via one-click launcher	CSV can be sensitive to concurrent manual edits

7. Process Simplification Impact

The automation consolidates multiple tools and manual lookups into one governed SOC interface. Analysts can process candidate IPs, validate risk, and complete approval documentation significantly faster with fewer transcription errors.